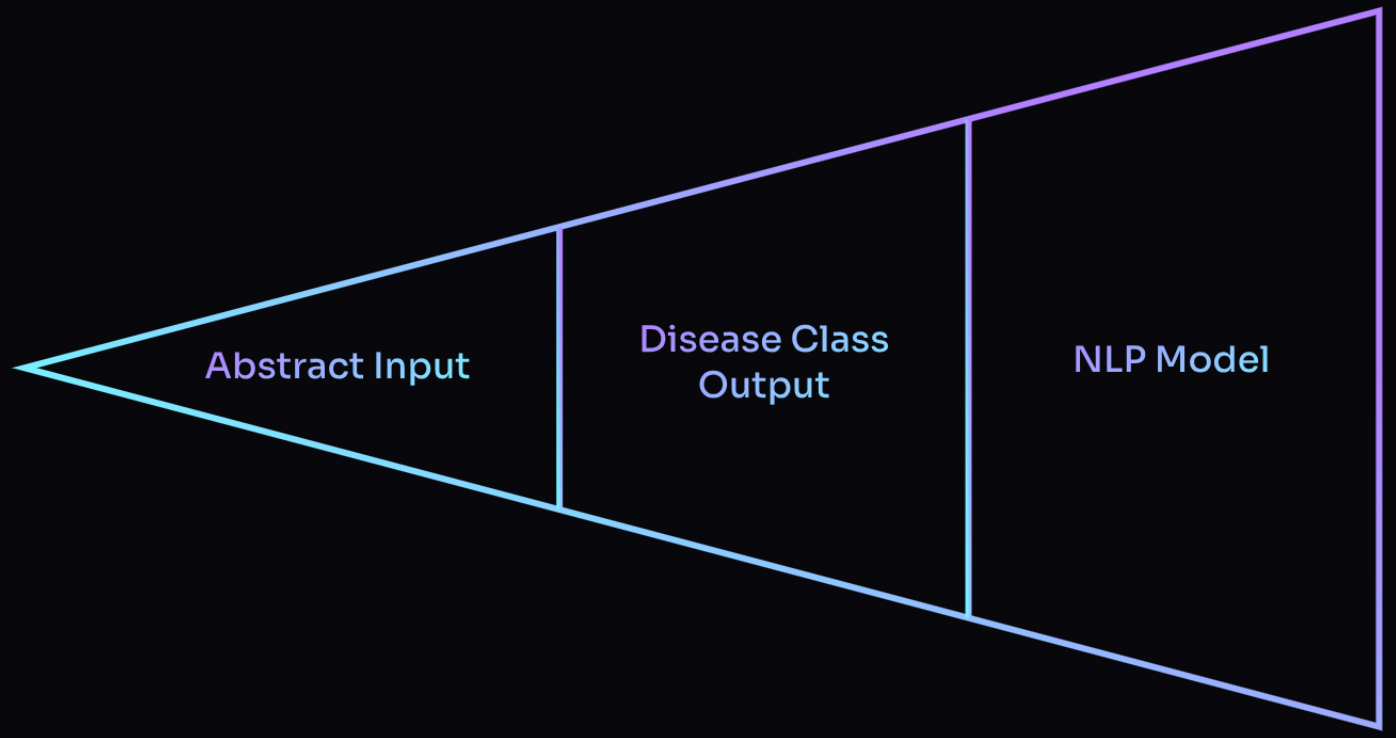
Team Ultron

Medical Abstract Disease Classification

NLP · BIOMEDICAL TEXT CLASSIFICATION · MULTI-CLASS CLASSIFICATION



Objective

Classify medical abstracts into 5 disease categories using classical ML, neural networks, and biomedical transformers.

Team Ultron

Members working across data engineering, model development, and evaluation design.

Dataset Audit & Cleaning

Real data work — not just model training. A rigorous pipeline to ensure zero leakage across splits.

Original Corpus

14,438 raw abstracts · 5 disease classes · Medical Abstracts Text Classification Corpus

Why This Dataset?

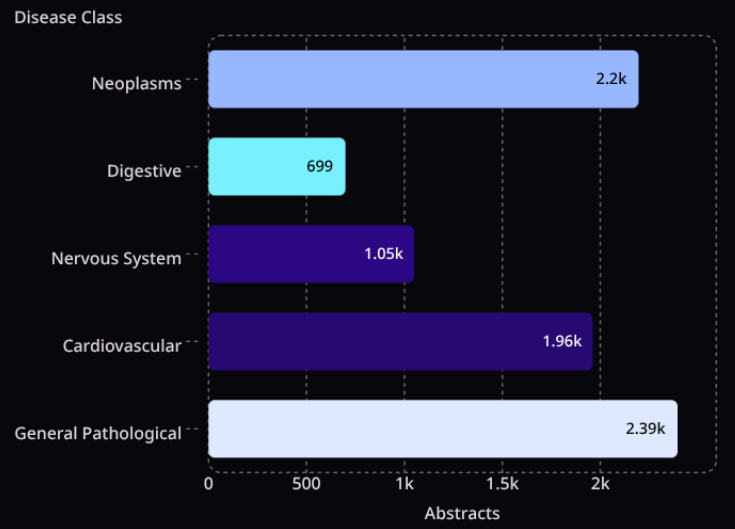
Prior drug-recall dataset dropped: heavily dominant class + repetitive data. This corpus offered a cleaner multi-class problem.

Final Clean Corpus

8,298 unique abstracts after removing 2,929 conflicting groups and 988 train/test overlaps.

Train	Val	Test
5,808	**1,245**	**1,245**

Stratified split · Seed = 42 · **Zero overlap** across final splits



Source: Schopf et al. (2023) — github.com/sebischair/Medical-Abstracts-TC-Corpus — DOI: 10.1145/3582768.3582795

Evaluation Strategy & Benchmark Design

Primary: Macro F1

Equal weight per class — accuracy alone misleads.

Supporting Metrics

Accuracy · Precision/Recall · Weighted F1 · Confusion Matrix

Methodology

All tuning on validation only. Test set locked until final eval.

Why accuracy misleads

Naive Bayes: **Accuracy ≈ 67.6%** but **Macro F1 = 0.4849**. High accuracy can hide poor minority-class performance.

Model Progression

Testing whether **model complexity** or **domain-specific pretraining** drives the biggest gains.

1

Classical

Naive Bayes · Logistic Regression · Linear SVM

2

Neural

TextCNN · BiLSTM

3

General Transformers

BERT-base · DistilBERT

4

Biomedical Transformers

BioBERT · BiomedBERT · DeBERTa-v3

BERT	BioBERT / BiomedBERT	DeBERTa-v3
General-purpose transformer architecture	BERT architecture pretrained on biomedical corpora	Newer transformer with disentangled attention

Experiments & Technical Decisions

Experiment	Result / Decision
TF-IDF representation	LR strongest w/ word unigrams; SVM strongest w/ word + char TF-IDF
Hyperparameter tuning	LR C=2 · SVM C=0.5 · NB alpha tuning
Naive Bayes tuning	Macro F1: **0.4849 → 0.7889**
TextCNN filters	Filter widths (2,3,4,5) strongest
BiLSTM padding	Fixed with packed sequences; tested class weighting
Token-length analysis	Only **1.36%** of BiomedBERT inputs exceed 512 tokens
DeBERTa memory	T4 OOM → batch 2 + gradient accumulation 4
Biomedical pretraining	BioBERT / BiomedBERT clearly outperform general BERT

Key Engineering Choices

Packed Sequences

Corrected BiLSTM padding for variable-length inputs

Gradient Accumulation

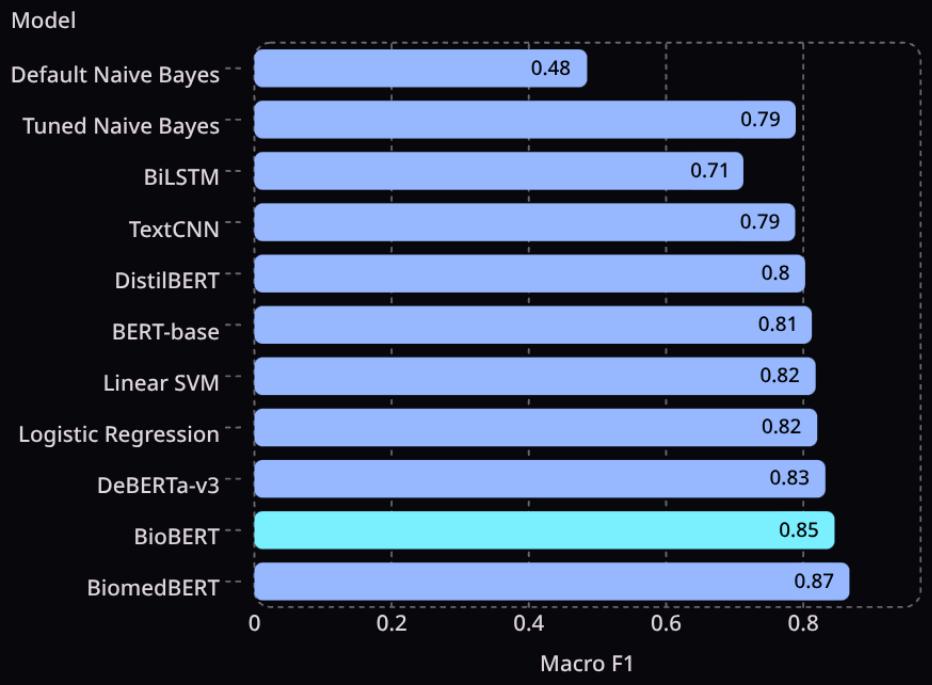
4 steps at batch size 2 to fit DeBERTa on T4 GPU

Class-Weighted Loss

Applied to address minority class under-representation

Benchmark Results

Ranked by validation Macro F1 — biomedical pretraining, not just scale, drove the top performance.



Key Insight

Biomedical pretraining mattered more than simply using a larger or more complex model.

BERT vs BiomedBERT

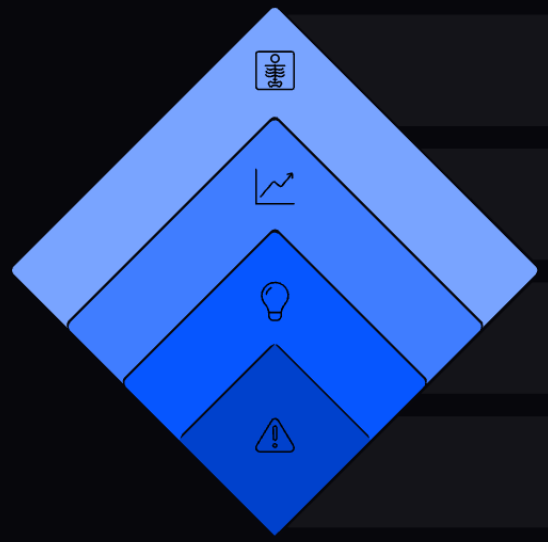
Same general parameter scale — **0.8124 → 0.8672**. Domain pretraining alone accounts for the jump.

Classical Baseline

Tuned Logistic Regression (**0.8203**) outperformed DistilBERT and matched general BERT — a strong baseline.

Error Analysis → Hierarchical Experiment

82–89% of errors across multiple models involved the **General Pathological Conditions** class — motivating a structural solution.



Decision
Reject hierarchical due to routing errors

Two-Stage Test
Binary then 4-class classification

Hypothesis
Separate General class first

Observation
General class causes most confusion

Model	Macro F1
Flat BiomedBERT	**0.8672** ✓ Selected
Combined Hierarchical	0.8599 — Rejected

Routing errors in Stage 1 cascaded through Stage 2, reducing overall performance. The flat model was retained.

Stage 1 Result

Specific vs. General binary split
Macro F1 = **0.8572**

Stage 2 Result

4 specific disease classes
Macro F1 = **0.9478**

Final Model Reliability & Efficiency

86.83%

Validation Accuracy

0.8672

Macro F1

97.06%

Mean Confidence

35.62

Abstracts / sec

T4 GPU · 28.07 ms/abstract

Calibration

ECE = **0.1041** — model is **overconfident**. Confidence scores should not be interpreted as exact probabilities.

Threshold Filtering

Confidence ⩾ 0.90 → coverage **92.61%** (1,153 / 1,245 accepted) · accuracy lifts to **89.68%**

Model Size

109,486,085 parameters — standard BERT-scale footprint

Inference Speed

Practical inference speed: **35.62 abstracts/sec** on T4

Final Locked Test Result

ONE-TIME EVALUATION · NO TUNING ON TEST SET

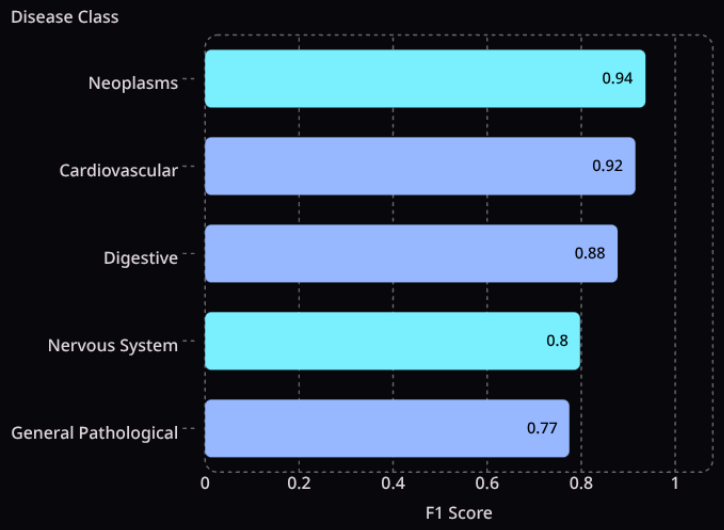
86.35%
Test Accuracy

0.8603
Test Macro F1

0.8563
Macro Precision

0.8651
Macro Recall

Per-Class F1 Scores



Validation → Test Gap

Val Macro F1: **0.8672**
Test Macro F1: **0.8603**
Δ = 0.0069 — small validation-to-test gap indicates stable generalization.

Hardest Class

General Pathological Conditions (F1 = **0.7746**) remains the most ambiguous — consistent with error analysis findings.

Weighted F1: **0.8623** — strong across all five disease classes.

Conclusions

1 Data Quality First

Auditing removed conflicts and leakage — essential before any modeling began.

2 Strong Classical Baselines

Tuned Logistic Regression and SVM remained competitive, outperforming several transformer variants.

3 Domain Pretraining > Scale

Biomedical-domain pretraining delivered the largest single performance jump across all experiments.

4 Final Model: BiomedBERT

Selected as the final model. Test Macro F1 = **0.8603** with stable validation-to-test generalization.

Limitation 1

General Pathological Conditions remains the most ambiguous class — inherently hard to separate.

Limitation 2

Confidence scores are overconfident (ECE = 0.1041) and should not be used as calibrated probabilities.